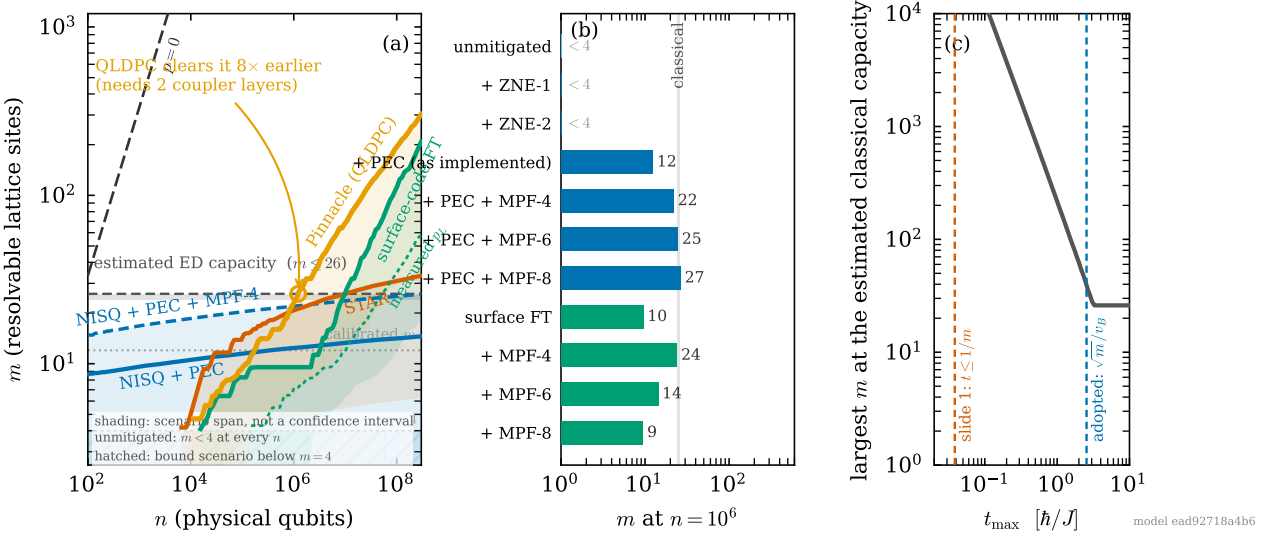

(a)
QLDPC clears it 8× earlier
(needs 2 coupler layers)
Pinnacle (QLDPC)
surface, code FT
measured
estimated ED capacity ($m \le 26$)
NISQ + PEC + MPF-4
NISQ + PEC
SHOR
calibrated m
shading: scenario span, not a confidence interval
unmitigated: $m < 4$ at every n
hatched: bound scenario below $m = 4$
m (resolvable lattice sites)
n (physical qubits)
(b)
unmitigated
+ ZNE-1
+ ZNE-2
+ PEC (as implemented)
+ PEC + MPF-4
+ PEC + MPF-6
+ PEC + MPF-8
surface FT
+ MPF-4
+ MPF-6
+ MPF-8
< 4
< 4
< 4
12
22
25
27
10
24
14
9
classical
m at $n = 10^6$
(c)
largest m at the estimated classical capacity
slice 1: $t \le 1/m$
adopted: $\sqrt{m}/v_B$
$t_{\max}$ $[\hbar/J]$
model ead92718a4b6